
Designing an audio amplifier using LM386 op-amp ic , the load is a speaker.

Course title : Electronics II

Course code : EEE 227



Department of Electrical and Electronic Engineering

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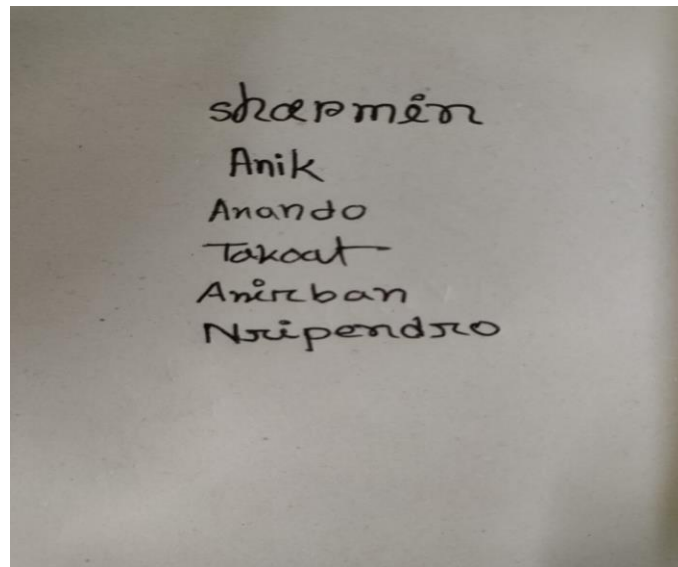
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Affidavit

We, Group No. 08, undersigned, therefore declare that the work included in this report is our original work, completed in compliance with the values of accountability, honesty, and integrity under the scientific guidance of Dr. Md. Rasedujjaman, Ph.D. The report, which complies with the charter on the fight against scientific plagiarism, was authored entirely by us.

Sylhet, January 20, 2024



A photograph of a piece of paper with handwritten signatures in black ink. The signatures are arranged vertically and read from top to bottom: 'sharmen', 'Anik', 'Anando', 'Takaat', 'Anirban', and 'Nrupendro'.

sharmen
Anik
Anando
Takaat
Anirban
Nrupendro

Aim:

- To design an audio amplifier using LM 386 IC where the load is a speaker. .

Introduction:

The design and implementation of a compact audio amplifier, a crucial part of audio systems, are examined in this project report. The amplifier uses the LM386 Operational Amplifier with an emphasis on efficiency and simplicity. An overview of the amplifier's design, salient characteristics, and useful considerations is given in this study.

Equipment required:

- Veroboard,
- Connecting wires,
- Op-Amp (LM386N-2)
- 9-12 V battery,
- Resistors: 1k ohms(2) ,
- Capacitors: 100(2) μ F, 0.1 μ F, 1000 μ F
- A speaker
- Multimeter.

Project Theory: Audio Amplifier using LM386N Operational Amplifier

The primary objective of this project is to design an audio amplifier circuit using the LM386N Operational Amplifier (Op-Amp). This versatile and widely used IC is known for its simplicity and effectiveness in audio applications.

- **Input Configuration:** Two resistors are strategically employed. Each stereo channel is connected to one resistor, and the junction of these resistors serves as the mono input to the LM386N Op-Amp. This simple resistor network combines the left and right audio signals, resulting in a mono audio output.
- **Coupling Capacitors:** To facilitate DC isolation and allow only the AC audio signal to pass through, 100 microfarad capacitors are coupled to pins 6 and 7 of the LM386N Op-Amp. These capacitors ensure that any DC bias in the input signal is

blocked, preventing potential damage to the amplifier and improving overall performance.

- **Filtering Capacitors:** In order to enhance audio signal quality and eliminate high-frequency distortion, two capacitors are strategically placed. A 1000 microfarad capacitor is positioned between the output of the LM386N Op-Amp and the speaker. This capacitor acts as a low-pass filter, attenuating high-frequency components and allowing only the audio signal to reach the speaker. Additionally, a 100-microfarad capacitor is connected from the output to the ground. This capacitor serves as a decoupling capacitor, filtering out any residual high-frequency noise or distortion before the signal is delivered to the load.
- **Overall Functionality:** The LM386N Op-Amp, configured in a non-inverting amplifier mode, amplifies the mono audio signal, providing sufficient gain to drive the connected speaker. The combination of input resistors and coupling and filtering capacitors ensures that the amplified audio signal is mono, free from DC bias, and devoid of unwanted high-frequency artifacts. In summary, this project capitalizes on the LM386N Op-Amp's capabilities to create a simple yet effective stereo to mono audio amplifier, demonstrating the integration of resistors and capacitors to shape the input, enhance signal quality, and deliver a clean and amplified audio output.

LM 386-2 IC:

The LM386N-2 is a compact audio amplifier IC. Notable features include a fixed voltage gain of 20, low quiescent power consumption, and a wide supply voltage range (4V to 12V). With internal output protection, it is ideal for applications with short circuits or excessive loads. The LM386N-2's simplicity, requiring minimal external components, makes it a popular choice for audio amplification projects, especially in portable devices. Its 8-pin DIP package and distinct gain setting differentiate it from other variants like the LM386N-1. Overall, the LM386N-2 is valued for its reliability and ease of integration in various audio applications.

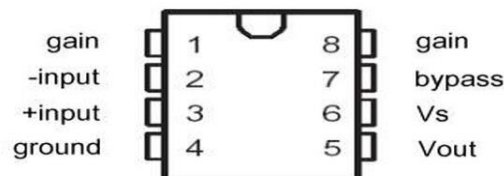


Fig : LM 386N-2

Design :

For the simplicity purpose we used 1k ohms registers to receive the inputs, and the capacitors we used are two 100 μF and One 0.1 μF and a 1000 μF .

The load we used in the project is a 4 ohm 10 watt speaker.

Our op amp is a LM386N-2 IC.

$$V_i = 0.2 \text{ V}$$

$$V_o = 2 \text{ V}$$

$$\text{Gain} = (2/.2)$$

$$= 20$$

Circuit diagram:

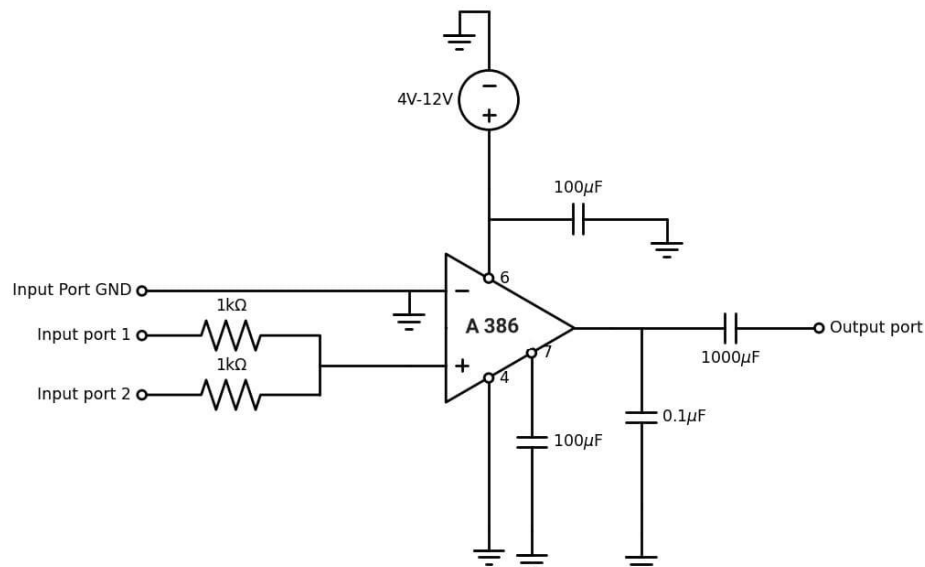


Fig : Project circuit diagram

Schemdraw code:

```
import schemdraw
```

```
import schemdraw . elements as elm
```

```
with schemdraw . Drawing ( ) as d :
```

```
    d += (op := elm.Opamp(leads=True)).label('A386', loc='center', ofst=0)
```

```
    d += elm.Line().left(d.unit * 1.5).at(op.in1)
```

```

d += elm.Dot(open=True).label('Input Port GND',loc='left')

d += elm.Ground().at(op.in1)

d += (L1:= elm.Line()).left(d.unit * 0.50).at(op.in2))

d += elm.Line('up').length(0.5)

d += elm.Resistor('left').label('1k $\Omega$ ')

d += elm.Dot(open=True).label('Input port 1',loc='left')

d += elm.Line('down',at=(L1,'end')).length(0.5)

d += elm.Resistor('left').label('1k $\Omega$ ')

d += elm.Dot(open=True).label('Input port 2',loc='left')

d +=elm.Line('down',at=(op.vs)).length(2.5)

d += elm.Ground()

d +=elm.Capacitor('down',at=(op.n1)).label('100 $\mu$ F',loc='bot')

d += elm.Ground()

d += elm.Line('up').length(1).at(op.vd)

d += (C1:= elm.Capacitor('rt').label('100 $\mu$ F'))

d += elm.Ground()

d += elm.Line('up',at=(C1,'start')).length(0.5)

d += elm.SourceV().reverse().label('4V-12V')

d += elm.Line('left').length(0.5)

d += elm.Ground()


d += elm.Dot(open=True).at(op.vd).label('6',loc='right')

d += elm.Dot(open=True).at(op.n1).label('7',loc='right')

d += elm.Dot(open=True).at(op.vs).label('4',loc='right')

```

```

d += (L2:= elm.Line('right',at=(op.out)).length(0.5))

d += elm.Capacitor('down',at=(L2,'end')).label('0.1$\mu$F',loc='bot')

d += elm.Line().length(0.45)

d += elm.Ground()

d += (Co:= elm.Capacitor('right',at=(L2,'end')).label('1000$\mu$F',loc='bot'))

d += elm.Dot(open=True).label('Output port',loc='right')

d.draw()

```

Project picture:

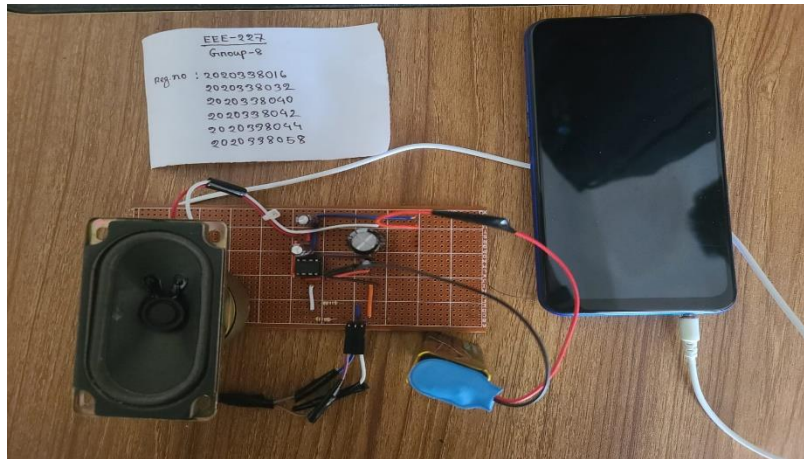


Fig : The Project

Simulation Circuit and Result:

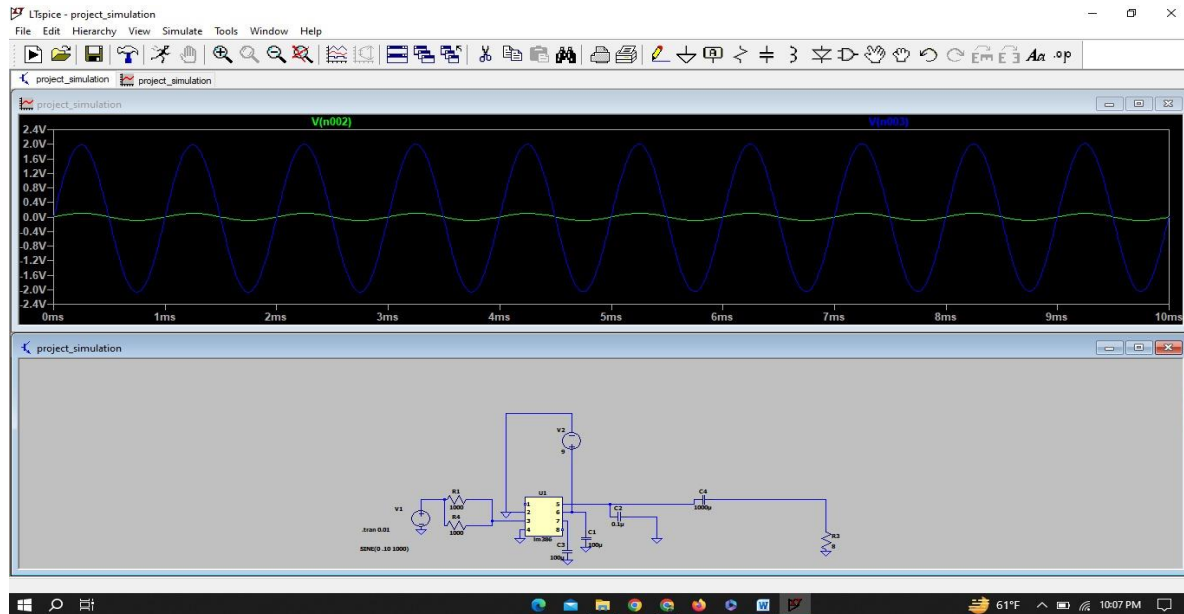


Fig : Project simulation and output.

Conclusion and Discussion:

The audio amplifier works perfectly under the rated load and wattage as we used an 4 Ω 10W speaker. But sometimes distorted sound as output can be heard because of bad input signal from the audio jack and also insufficient supply of DC power . But with constant supply and a good input solution we can solve the problem completely.